GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	**FEED for Gas Gathering Pipelines & Associated Infrastructure**
Client	**Brass Fertilizer and Petrochemical Company Limited**
Originating Company	**Epic Atlantic Limited**
Document Title	**Critical Line List-BBOU**
Document Number	**EA-BFPCL-GPMC-GGPL-FEED-MPP-LST-011**
Document Revision	**R01**
Document Status	Issued For Review
Originator	Joseph Obinna
Security Classification	Restricted
Issue Date	**1 Aug 2022**

Critcal Line List-BBOU

R01	01.08.22	Issued for R eview	J.Obinna	E. Nwachukwu	A. Duru	
Rev #	**Issue Date**	**Status Description**	**Originator**	**Reviewer**	**Approver (Epic)**	**Approver (BFPCL)**

Revision History

Rev No.	Date	Description of Revision
R01	1.08.22	Issued for Review

Revision Philosophy:

a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.

b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.).
for the subsequent project phases, post FEED, it is expected that:

c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.

d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)

e. All approved “As-Built” documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the “As-Built” document to Z02).

GENERAL NOTES

1. In accordance with DEP 31.38.01.26-Gen, the following in-plant process lines shall be considered for a comprehensive piping stress analysis using Industry recognised Stress analysis software. Bentley AutoPIPE CONNECT v11.2 will be used.

- Process lines size DN 80 and above (NPS 3 and above) subjected to slug flow or water hammer

- Process lines size of size DN 80 and above (NPS 3 and above) subjected to two phase flow

- Process lines size DN 50 (NPS 2) and above subjected to relief loads;
- Process lines size DN 300 (NPS 12) with design temperatures from 40 °C (100 °F) to 175 °C (350 °F)
- Blowdown and flare header or vent piping, but not atmospheric vent piping
- Lines having substantial concentrated loads such as valves, fittings, unsupported vertical risers and branches
- Pipe size DN 500 (NPS 20) and larger

2. The formal computerised analysis shall provide report in chart methods that provide an evaluation of the forces, moments and stresses caused by displacement strains.
3. The integrity and flexibility of lines that are not on the critical line list (“non-critical”) shall be guarranted and shall be properly supported.
4. All critical lines shall be identified as such on the project line-list.
6. Stress analysis of the main product pipeline section, from pig reciever to pig launcher shall be conducted separately by pipeline discipline When full survey information/Data on the Right of Way (ROW) is available.

S/No.	LINE NUMBER DESCRIPTION						LINE NUMBER	MATERIAL	OPERATING COND.		DESIGN CONDITION				LINE ORIGIN		LINE DESTINATION		
	System	LOCATION CODE	NOMINAL LINE SIZE	LINE SERVICE DESIGNATION	SERIAL NUMBER	PIPING CLASS			TEMP. (°C)	PRES. (bar.g)	DES. TEMP (°C)	DES. PRES. (bar.g)	HYDROTEST PRESS. (bar.g)	Service	ITEM OR LINE NO. FROM	P&ID DRAWING NO	ITEM OR LINE NO. TO	P&ID DRAWING NO	REMARKS
1	Process System	BBOU	24"	P	001	C70	BBOU 24" P 001 C70	ASME B36.10/B36.19, ASTM A790 S31803/ASTM A928 S31803 CL 1	58	100	-29 / 100	110	165	Multiphase(Gas+condensate)	TP-101	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	SDV 1001	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	
2		BBOU	24"	P	003	C03	BBOU 24" P 003 C03	ASME B36.10, API SPEC 5L GRADE X60 PSL 2	58	100	-29 / 100	110	165	Multiphase(Gas+condensate)	SDV 1001	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	BBOU-24"-P-008-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02	
2		BBOU	4"	P	004	C03	BBOU 4" P 004 C03	ASME B36.10, API SPEC 5L GRADE B PSL 2	58	100	-29 / 100	110	165	Multiphase(Gas+condensate)	BBOU-24"-P-003-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	BBOU-4"-B-002-B03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	
3		BBOU	6"	P	005	C03	BBOU 6" P 005 C03	ASME B36.10, API SPEC 5L GRADE B PSL 2	58	100	-29 / 100	110	165	Multiphase(Gas+condensate)	BBOU-24"-P-003-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	BBOU-6"-P-006-C03 BBOU-6"-P-007-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	
3		BBOU	24"	P	012	CL900	BBOU 24"P 012 CL900	ASME B36.10, API SPEC 5L GRADE X70 PSL 2	98	100	-29 / 100	110	165	Multiphase(Gas+condensate)	OPUGBENE MANIFOLD PLATFORM	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	SDV 1003	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	
4		BBOU	24"	P	012	CL900	BBOU 24" P 012 CL900	ASME B36.10, API SPEC 5L GRADE X70 PSL 2	98	100	-29 / 100	110	165	Multiphase(Gas+condensate)	SDV 1003	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	BBOU-A-100 PIG RECEIVER)	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	
4		BBOU	24"	P	013	CL900	BBOU 24" P 013 CL900	ASME B36.10, API SPEC 5L GRADE X70 PSL 2	98	100	-29 / 100	110	165	Multiphase(Gas+condensate)	BBOU-24"-PL-012-X70-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	TP 101	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	
5		BBOU	4"	P	015	C03	BBOU 4" P 015 C03	ASME B36.10, API SPEC 5L GRADE B PSL 2	98	100	-29 / 100	110	165	Multiphase(Gas+condensate)	4" MV1070	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	BBOU-8"-P-014-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	
5		BBOU	8"	P	14	C03	BBOU 8" P 014 C03	ASME B36.10, API SPEC 5L GRADE B PSL 2	98	100	-29 / 100	110	165	Multiphase(Gas+condensate)	BBOU-A-100 PIG RECEIVER)	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	BBOU-24"-P-0013-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	
6		BBOU	4"	P	016	C03	BBOU 4" P 016 C03	ASME B36.10, API SPEC 5L GRADE B PSL 2	98	100	-29 / 100	110	165	Multiphase(Gas+condensate)	4" MV1072	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	PSV 1003	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	
6		BBOU	24"	P	003	C03	BBOU 24" P 003 C03	ASME B36.10, API SPEC 5L GRADE X60 PSL 2	98	100	-29 / 100	110	165	Multiphase(Gas+condensate)	BBOU MANFOLD HEADER OPUGBENE-BBOU PIPELINE PIG RECEIVER	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	24" MV 1022	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02	
7		BBOU	6"	P	011	C03	BBOU 6" P 011 C03	ASME B36.10, API SPEC 5L GRADE B PSL 2	98	100	-29 / 100	110	165	Multiphase(Gas+condensate)	BBOU-A-200 (PIG LAUNCHER)	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02	PSV 1004	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02	
7		BBOU	8"	P	009	C03	BBOU 8" P 009 C03	ASME B36.10, API SPEC 5L GRADE B PSL 2	98	100	-29 / 100	110	165	Multiphase(Gas+condensate)	BBOU-24"-P-'008-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02	BBOU-A-200 (PIG LAUNCHER)	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02	
8		BBOU	4"	P	010	C03	BBOU 4" P 010 C03	ASME B36.10, API SPEC 5L GRADE B PSL 2	98	100	-29 / 100	110	165	Multiphase(Gas+condensate)	BBOU-8"-P-'009-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02	BBOU-A-200 (PIG LAUNCHER)	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02	
8		BBOU	24"	P	012	CL900	BBOU 24" P 012 CL900	ASME B36.10, API SPEC 5L GRADE X70 PSL 2	98	100	-29 / 100	110	165	Multiphase(Gas+condensate)	BBOU-A-200 (PIG LAUNCHER)	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02	SDV 1002	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02	
9		BBOU	24"	P	0XX	CL900	BBOU 24" P 0XX CL900	ASME B36.10, API SPEC 5L GRADE X70 PSL 2	98	100	-29 / 100	110	165	Multiphase(Gas+condensate)	SDV 1002	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02	ELEPA MANIFOLD	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_04	
9	Pressure Relief & Blowdown System	BBOU	6"	B	003	B03	BBOU 6" B 003 B03	ASME B36.10, API SPEC 5L GRADE B PSL 2	50	10	-29 / 100	17.7	26.55	Vent Gas	PSV 1001	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	BBOU-6"-B-001-B03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	
10		BBOU	6"	B	004	B03	BBOU 6" B 004 B03	ASME B36.10, API SPEC 5L GRADE B PSL 2	50	10	-29 / 100	17.7	26.55	Vent Gas	PSV 1002	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	BBOU-6"-B-001-B03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	
10		BBOU	6"	B	001	B03	BBOU 6" B 001 B03	ASME B36.10, API SPEC 5L GRADE B PSL 2	50	10	-29 / 100	17.7	26.55	Vent Gas	BBOU-4"-B-002-B03 BBOU-6"-B-003-B03 BBOU-6"-B-004-B03 TP-102	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	BBOU-ELEPA PIPELINE PIG LAUNCHER	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02	
11		BBOU	6"	B	005	B03	BBOU 6" B 005 B03	ASME B36.10, API SPEC 5L GRADE B PSL 2	50	10	-29 / 100	17.7	26.55	Vent Gas	PSV 1003	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	TP 102	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	
11		BBOU	6"	B	001	B03	BBOU 6" B 001 B03	ASME B36.10, API SPEC 5L GRADE B PSL 2	50	10	-29 / 100	17.7	26.55	Vent Gas	BBOU MANFOLD HEADER OPUGBENE-BBOU PIPELINE PIG RECEIVER	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	BBOU VENT & DRAIN SYSTEM	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_03	
12		BBOU	X"	B	007	B03	BBOU X" B 007 B03	ASME B36.10, API SPEC 5L GRADE B PSL 2	50	10	-29 / 100	17.7	26.55	Vent Gas	PSV 1004	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02	BBOU-6"-B-001-B03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02	
12		BBOU	6"	B	001	B03	BBOU 6" B 001 B03	ASME B36.10, API SPEC 5L GRADE B PSL 2	50	10	-29 / 100	17.7	26.55	Vent Gas	BBOU-ELEPA PIG LAUNCHER (BBOU-A-200)	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02	BBOU-CDT-001 (CLOSED DRAIN TANK)	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_03	
13		BBOU	6"	B	007	B03	BBOU 6" B 007 B03	ASME B36.10, API SPEC 5L GRADE B PSL 2	50	10	-29 / 100	17.7	26.55	Vent Gas	BBOU-CDT-001 (CLOSED DRAIN TANK)	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_03	VENT STACK	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_03	
13	Cloesd & Open Drain System	BBOU	4"	D	002	C03	BBOU 4" D 002 C03	ASME B36.10, API SPEC 5L GRADE B PSL 2	50	100	-29 / 100	110	165	Drain	BBOU-A-100 PIG RECEIVER)	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	TP 103	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	
14		BBOU	4"	D	004	B03	BBOU 4" D 004 B03	ASME B36.10, API SPEC 5L GRADE B PSL 2	50	10	-29 / 100	17.7	26.55	Drain	TP 103	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	BBOU-ELEPA PIPELINE PIG LAUNCHER	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02	
14		BBOU	2"	D	005	B03	BBOU 2" D 005 B03	ASME B36.10, API SPEC 5L GRADE B PSL 2	50	10	-29 / 100	17.7	26.55	Drain	DRIP PAN	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	TP 104	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	
15		BBOU	2"	D	003	B03	BBOU 2" D 003 B03	ASME B36.10, API SPEC 5L GRADE B PSL 2	50	10	-29 / 100	17.7	26.55	Drain	TP 104	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_01	BBOU-ELEPA PIPELINE PIG LAUNCHER	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02	
15		BBOU	4"	D	006	C03	BBOU 4" D 006 C03	ASME B36.10, API SPEC 5L GRADE B PSL 2	50	100	-29 / 100	110	165	Drain	BBOU-A-200 (PIG LAUNCHER)	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02	BBOU-4"-D-'007-B03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02	
16		BBOU	4"	D	007	B03	BBOU 4" D 007 B03	ASME B36.10, API SPEC 5L GRADE B PSL 2	50	10	-29 / 100	17.7	26.55	Drain	BBOU-4"-D-'006-C03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02	BBOU-4"-D-'001-B03	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02	

16		BBOU	4"	D	001	B03	BBOU 4" D 001 B03	ASME B36.10, API SPEC 5L GRADE B PSL 2	50	10	-29 / 100	17.7	26.55	Drain	BBOU MANFOLD HEADER OPUGBENE-BBOU PIPELINE PIG RECEIVER	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02	BBOU VENT & DRAIN SYSTEM	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_03	
17		BBOU	4"	D	001	B03	BBOU 4" D 001 B03	ASME B36.10, API SPEC 5L GRADE B PSL 2	50	10	-29 / 100	17.7	26.55	Drain	BBOU-ELEPA PIG LAUNCHER (BBOU-A-200)	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_02	BBOU-CDT-001 (CLOSED DRAIN TANK)	EA-BFPCL-GPMC-GGPL-FEED-PSE-DWG-003_03	